Exercise Sheet

Task 1 – Graph Structures

The connection structure of a multiprocessor is given as follows: The multiprocessor consists of n processor nodes $P_0, P_1, ..., P_{n-1}$ where each node P_i is connected to all nodes Pm where either $m = (i-1) \mod 10$ or $m = (i+1) \mod 10$

a) Draw the connection structure for $n = 10$. What type of structure is this?

b) A path is a list of edges $(\{v_1, v_2\}, ..., \{v_{a-1}, v_a\})$ where v_1 is the start node and v_a is the end node. State two paths on which a message can be sent from P_1 to P_4, how many paths for this connection are there in total?

c) In a similar structure of n nodes, all nodes $P_0, P_1, ..., P_{n-1}$ are connected in the following fashion: Each node P_i is connected to all nodes P_m where either $m = (i-2) \mod 10$ or $m = (i+2) \mod 10$
Draw the corresponding connection structure for $n = 10$

d) A connection component is a graph of a subset of nodes, where each of those nodes can be reached from all other nodes within the subset. How many connection components does the structure from c) consist of?

e) Derive general formulae on degree and number of connections depending on $|V|$ for the structure from c). Explain why $Diameter(G) = \infty$

Task 2 – Hypercubes

a) For a hypercube where $n = 4$, decide the values of its diameter, degree and how many connections there will be.

b) Draw this hypercube.